No.

Date

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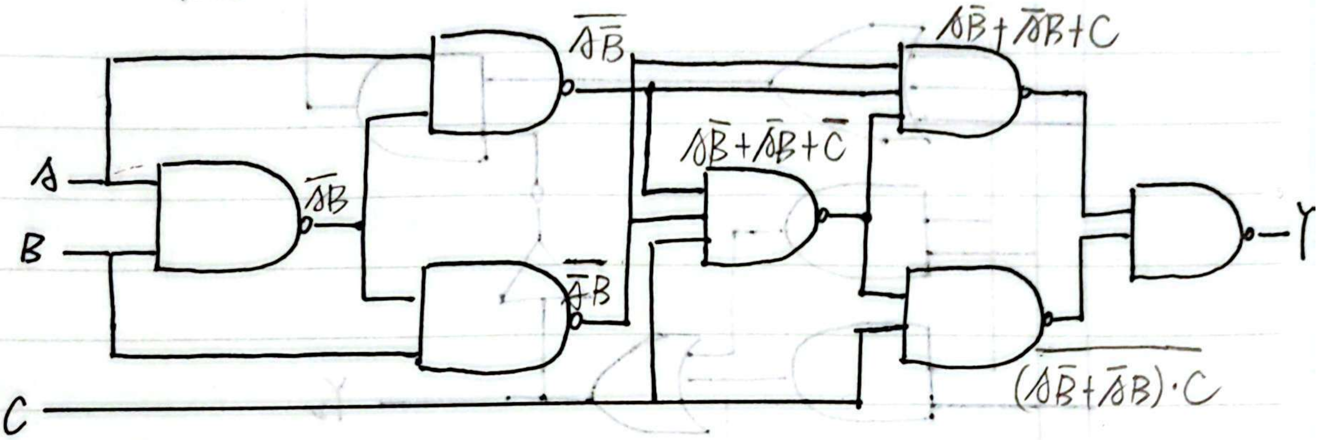
VLSI - 第7周作业

1. (1)



$$\overline{\overline{A\bar B}\cdot\overline{\bar A B}\cdot C} = A\bar B+\bar A B+\bar C$$

$$\overline{\overline{A\bar B}\cdot\overline{\bar A B}\cdot(A\bar B+\bar A B+\bar C)} = A\bar B+\bar A B+C$$

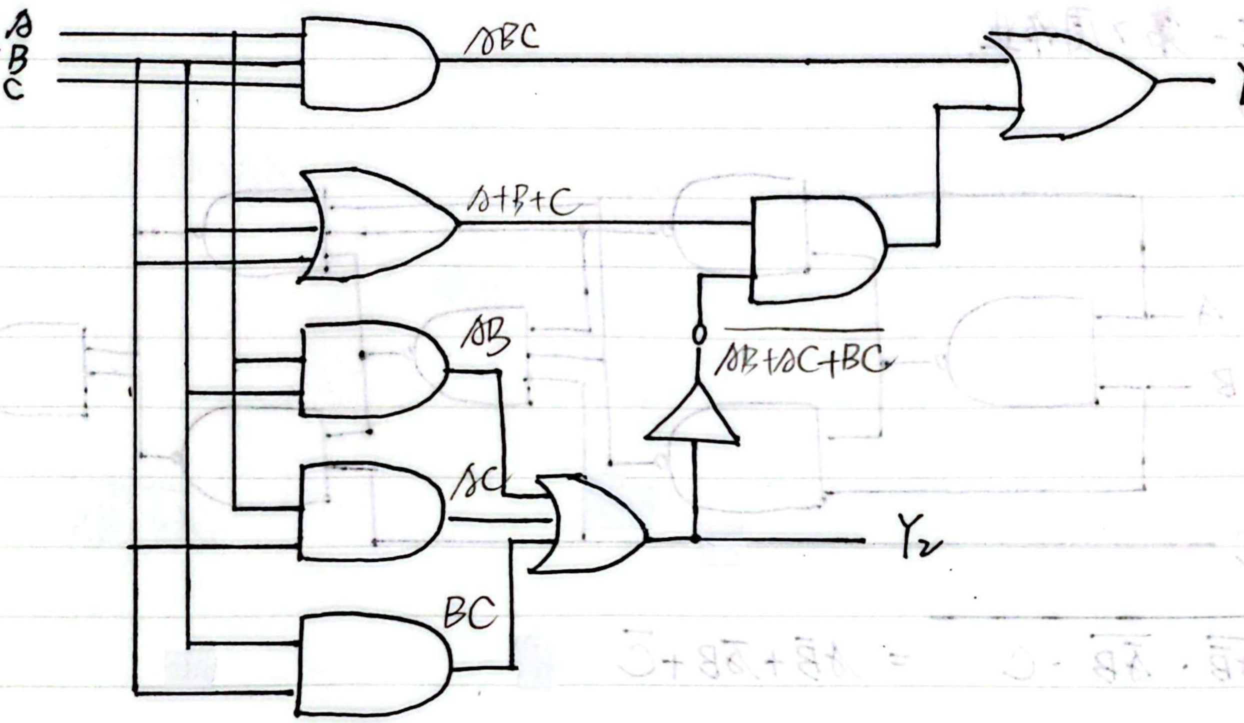
$$\therefore Y=\overline{(A\bar B+\bar A B+C)\cdot\overline{(A\bar B+\bar A B)\cdot C}} = \overline{A\bar B+\bar A B}\cdot\bar C+(A\bar B+\bar A B)\cdot C$$
$$=(A\oplus B)\odot C$$

A	B	C	Y
0	0	0	1
0	0	1	0
0	1	0	0
1	0	0	0
0	1	1	1
1	0	1	1
1	1	0	1
1	1	1	0

[注]. $(A\oplus B)\odot C=(A\odot B)\oplus C$

功能：判断 ABC 是否有偶数个1.

(2).



$Y_1 = A \oplus B \oplus C$

$Y_2 = AB + AC + BC$

A	B	C	Y_1	Y_2
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
1	0	0	1	0
0	1	1	0	1
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

功能. Y_1为 ~~全加~~ 2个一位全加器的本位

Y_2为 2个一位全加器的进位.

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2. 由图可知

$$Y = \overline{ABS_0} \cdot \overline{A'BS_1} \cdot \overline{AB'S_2} \cdot \overline{A'B'S_3}.$$

S_0	S_1	S_2	S_3	Y
0	0	0	0	1
0	0	0	1	$A+B$
0	0	1	0	$\overline{A}+B$
0	1	0	0	$A+\overline{B}$
1	0	0	0	$\overline{A}+\overline{B}$.
0	0	1	1	B
0	1	0	1	A
1	0	0	1	$A \oplus B$
0	1	1	0	$AB+\overline{A}\,\overline{B}$
1	0	1	0	$\overline{A}$
1	1	0	0	$\overline{B}$
0	1	1	1	AB
1	0	1	1	$\overline{A}B$
1	1	0	1	$A\overline{B}$
1	1	1	0	$\overline{A}\,\overline{B}$
1	1	1	1	0

3. 列出真值表

A	B	C	Ms	ML
0	0	0	0	0
0	0	1	1	0
0	1	0	X	X
1	0	0	X	X
0	1	1	0	1
1	0	1	X	X
1	1	0	X	X
1	1	1	1	1

(打X表示违法输入).

对于Ms

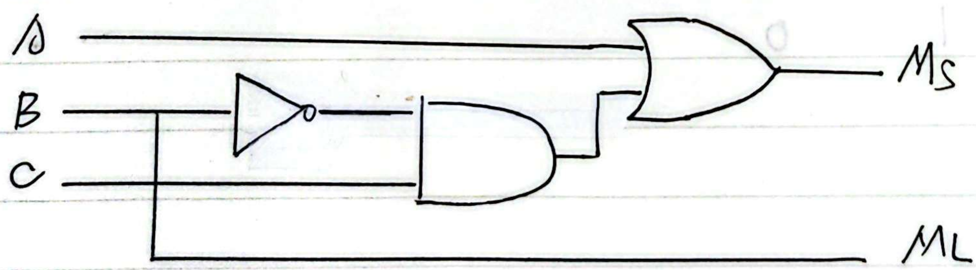
C \ AB	00	01	11	10
1	1	0	1	X
0	0	X	X	X

对于ML

C \ AB	00	01	11	10
1	0	1	1	X
0	0	X	X	X

令 $M_S = \cancel{\bar{A}\bar{B}C + ABC = (A \odot B)\cdot C}\ A + \bar{B}C$

$M_L = B$



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4. $Y_1 = AC = ABC + A\bar{B}C$ "$=7$" + "$=5$"

$Y_2 = A'B'C + AB'C' + BC = A'B'C + AB'C' + A'BC + ABC$ "$=1$" + "$=4$" + "$=3$" + "$=7$"

$Y_3 = B'C' + ABC' = A'B'C' + AB'C' + ABC'$ "$=0$" + "$=4$" + "$=6$"